WORKED SOLUTION

STEP III 2005, Question 6

Pure • Solving Cubic Equations

STEP III 2005, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q6

The key idea

The identity $4\cosh^3 y - 3\cosh y = \cosh 3y$ has exactly the shape of a "depressed" cubic $x^3 - 3bx$, once we scale x by the right constant. So a cubic with no x^2 term can be solved by matching it to the triple-angle formula, and the arcosh logarithm then turns the hyperbolic answer back into cube roots. Once one root is known, factorising gives a quadratic for the other two, and the cube roots of unity tidy them up.

Solution

The hyperbolic root

Put $x = 2a\cosh\frac{1}{3}T$. Then

$$\begin{aligned} x^3 - 3a^2x &= 8a^3\cosh^3\tfrac{1}{3}T - 6a^3\cosh\tfrac{1}{3}T \\ &= 2a^3(4\cosh^3\tfrac{1}{3}T - 3\cosh\tfrac{1}{3}T) \\ &= 2a^3\cosh T, \end{aligned}$$

using the given identity with $y = \frac{1}{3}T$. So $2a\cosh\frac{1}{3}T$ satisfies the equation, as required.

One root of $x^3 - 3bx = 2c$

Suppose first that $c > 0$. Since $b > 0$ we may take $a = \sqrt{b} > 0$, so that $3a^2 = 3b$, and we want $2a^3\cosh T = 2c$, that is,

$$\cosh T = \frac{c}{b^{3/2}}.$$

This needs $c/b^{3/2} \geq 1$, which is exactly where the condition is used: $c^2 \geq b^3$ with $c > 0$ gives $c \geq b^{3/2}$. So we may take

$$T = \operatorname{arcosh}\frac{c}{b^{3/2}} = \ln\left(\frac{c}{b^{3/2}} + \sqrt{\frac{c^2}{b^3} - 1}\right) = \ln\frac{c + \sqrt{c^2 - b^3}}{b^{3/2}}.$$

Hence

$$e^{T/3} = \frac{(c+\sqrt{c^2-b^3})^{1/3}}{b^{1/2}} = \frac{u}{\sqrt{b}}, \qquad e^{-T/3} = \frac{\sqrt{b}}{u},$$

and by the first part one root is

$$2\sqrt{b}\cosh\tfrac{1}{3}T = \sqrt{b}\left(\frac{u}{\sqrt{b}} + \frac{\sqrt{b}}{u}\right) = u + \frac{b}{u}.$$

Ask yourself What if $c < 0$? Then $c/b^{3/2} \le -1$ and no real T has that cosh.

Replace x by $-x$: the equation becomes $x^3 - 3bx = -2c$ with $-c > 0$, so by the case just done it has the root $u' + b/u'$ where $u' = (-c + \sqrt{c^2-b^3})^{1/3}$, and $-(u' + b/u')$ is a root of the original equation. Now

$$u^3 u'^3 = (\sqrt{c^2-b^3}+c)(\sqrt{c^2-b^3}-c) = (c^2-b^3) - c^2 = -b^3,$$

so, taking real cube roots, $u' = -b/u$ and $-(u' + b/u') = u + b/u$. So the formula holds for either sign of c. (Note $u \neq 0$ in both cases, since $\sqrt{c^2-b^3} < |c|$ when $b > 0$.)

The other two roots

Write $r = u + \frac{b}{u}$. Since r is a root, $x - r$ is a factor, and comparing coefficients in

$$x^3 - 3bx - 2c = (x-r)(x^2 + rx + q) = x^3 + (q - r^2)x - rq$$

(the x^2 terms cancel automatically) forces $q = r^2 - 3b$. Expanding,

$$q = u^2 + 2b + \frac{b^2}{u^2} - 3b = u^2 + \frac{b^2}{u^2} - b,$$

so the other two roots satisfy

$$x^2 + \left(u + \frac{b}{u}\right)x + u^2 + \frac{b^2}{u^2} - b = 0,$$

as required. (As a consistency check, the constant term also matches: $rq = u^3 + b^3/u^3$, and $b^3/u^3 = b^3/(c + \sqrt{c^2-b^3}) = c - \sqrt{c^2-b^3}$ after rationalising, so $rq = 2c$.)

To solve the quadratic, write $v = b/u$, so $uv = b$ and the quadratic is

$$x^2 + (u+v)x + (u^2 - uv + v^2) = 0.$$

Now $\omega^3 = 1$ and $1 + \omega + \omega^2 = 0$. Try $x_1 = \omega u + \omega^2 v$ and $x_2 = \omega^2 u + \omega v$:

$$x_1 + x_2 = (\omega + \omega^2)(u+v) = -(u+v),$$
$$x_1 x_2 = \omega^3 u^2 + \omega^2 uv + \omega^4 uv + \omega^3 v^2 = u^2 + (\omega^2 + \omega)uv + v^2 = u^2 - uv + v^2.$$

These are the sum and product of the roots of the quadratic, so its roots are

$$x = \omega u + \omega^2 \frac{b}{u} \qquad \text{and} \qquad x = \omega^2 u + \omega \frac{b}{u}.$$

Since $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$ and $\omega^2 = -\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$, these can also be written $-\frac{1}{2}(u + \frac{b}{u}) \pm \frac{\sqrt{3}}{2}(u - \frac{b}{u})\mathrm{i}$.

Solving $x^3 - 6x = 6$

This is $x^3 - 3bx = 2c$ with $b = 2$, $c = 3$, and $c^2 = 9 \geq 8 = b^3 > 0$, so the results apply. Here

$$u = (3 + \sqrt{9-8})^{1/3} = 4^{1/3} = 2^{2/3}, \qquad \frac{b}{u} = \frac{2}{2^{2/3}} = 2^{1/3}.$$

So the roots are

$$x = 2^{2/3} + 2^{1/3}, \qquad x = -\tfrac{1}{2}(2^{2/3} + 2^{1/3}) \pm \tfrac{\sqrt{3}}{2}(2^{2/3} - 2^{1/3})\mathrm{i},$$

the complex pair being $\omega\, 2^{2/3} + \omega^2\, 2^{1/3}$ and its conjugate.

Checks

- The real root is $2^{2/3} + 2^{1/3} \approx 2.847322$, and $2.847322^3 - 6(2.847322) \approx 23.083930 - 17.083932 \approx 6.0000$.
- The three roots should sum to 0 (no x^2 term): $(u+v) - (u+v) = 0$. Their product should be 6: $r(u^2 - uv + v^2) = u^3 + v^3 = 4 + 2 = 6$.
- If $c^2 = b^3$ then $u = \sqrt{b} = b/u$, and the two complex roots both become $-\sqrt{b}$, a repeated root, as $x^3 - 3bx - 2b^{3/2} = (x - 2\sqrt{b})(x + \sqrt{b})^2$ confirms.

Common slips

- Forgetting that $\cosh T \geq 1$, so the hyperbolic argument only works directly for $c > 0$; the case $c < 0$ needs the substitution $x \to -x$.
- Writing the complex roots as $\omega u + \omega \frac{b}{u}$: the product of the two cube roots must stay equal to b, so the powers of ω must multiply to 1.

Beyond STEP

This is Cardano's formula for the cubic, and the form $u + b/u$ with u^3 a root of a quadratic is how it was first found in the sixteenth century. When $c^2 < b^3$ the square root is imaginary even though all three roots are real (the "irreducible case"), and the trigonometric version $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ takes over. Why no formula of this kind exists for the general quintic is the story of Galois theory; Allenby, *Rings, Fields and Groups*, is a good way in.

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